

Detector Ranking is Not Architecture-Invariant under Medical Covariate Shift

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Abstract

Post-hoc out-of-distribution (OOD) detectors are almost always ranked on a single backbone. **Holding a medical covariate split fixed, logit-based scores reorder substantially with the backbone while feature-space scores (Mahalanobis, k NN) stay near the top throughout.** On Camelyon17 hospital-2 (WILDS), Kendall’s $W = 0.694$ across eight CNN families and seven scores (seed 42). MSP AUROC spans 0.515 (ResNet-50) to 0.883 (DenseNet-121)—a 0.37 swing, as large as Mahalanobis–MSP on that ResNet. Copying ResNet-50’s best logit score (ReAct) onto ConvNeXt costs 0.185 AUROC versus ConvNeXt’s own best logit. On the same labeled OOD test set, the AUROC-highest backbone has coverage@risk10% = 0.078 while the coverage@risk-highest backbone reaches 0.904: AUROC is not a reliable proxy for coverage@risk. Feature-space scores occupy ranks 1–2 on every official backbone. Skin ISIC→PAD does not reproduce the MSP jump ($W = 0.791$). AUROC versus coverage@risk disagreement *flips sign* between Camelyon and MIDOG. A pre-registered ViT-B/16 MSP of 0.884 met a locked threshold of 0.6947. The protocol: do not copy an OOD ranking across backbones, and do not pick a backbone for triage from an AUROC table.

Keywords: out-of-distribution detection, domain shift, histopathology, dermoscopy, benchmark evaluation, model selection

1. Introduction

We test one hypothesis: *under a fixed medical covariate shift, post-hoc detector ranking is an architecture property.* On Camelyon17 hospital-2, ResNet-50 MSP AUROC is 0.515 while DenseNet-121 is 0.883 (Figure 1a). A lab that copies an OpenMIBOOD-style ResNet-50 recipe and a lab that uses DenseNet are not ranking the same detectors; and, measured on the same labeled test set, AUROC order does not predict coverage@risk order, so an AUROC table is not a safe guide to triage performance—the AUROC-best and coverage@risk-best backbones differ by 0.826 coverage at a 10% risk budget (Figure 1b).

That hypothesis is *not* that ranking was previously assumed architecture-invariant. On natural-image semantic OOD, ranking already moves with architecture (Zhang et al., 2023; Szyk et al., 2023; Datko et al., 2026; Claros Olivares and Brockmeier, 2025). OpenMIBOOD (Gutbrod et al., 2025) already showed that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores under a frozen classifier per domain. The orthogonal question is: **holding one medical covariate-shift dataset fixed, how much does ranking move when only the backbone changes?**

The rest of the paper is that test, in order. Camelyon discovers the shift: Kendall $W = 0.694$ and a backbone-dependent MSP jump. Five seeds verify the DenseNet MSP

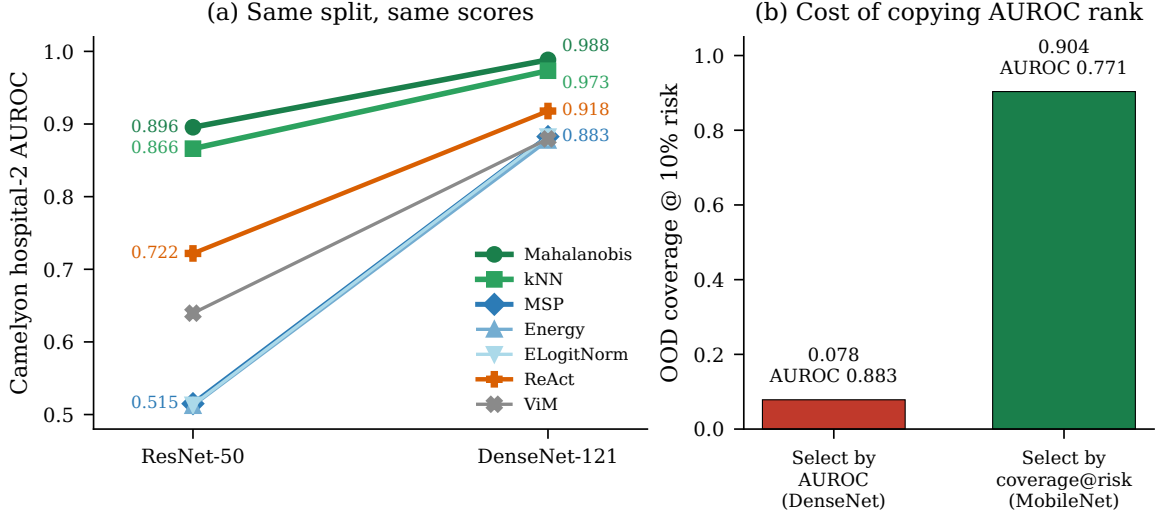


Figure 1: **The claim in one figure.** (a) Same Camelyon hospital-2 split, same seven scores: switching ResNet-50→DenseNet-121 lifts logit AUROC by ~ 0.37 while Mahalanobis and k NN stay on top. (b) On the same labeled test set, the AUROC-highest backbone (DenseNet) has coverage@risk 10% = 0.078 while the coverage@risk-highest backbone (MobileNet) reaches 0.904: AUROC rank and coverage rank disagree.

jump (5/5). Matched-recipe training asks whether that jump is architecture or optimizer. Skin falsifies a universal MSP-jump law, while feature ranks stay 1–2. MIDOG is the deployment readout: the *sign* of AUROC versus coverage@risk disagreement flips. A pre-registered ViT-B/16 tests whether the Camelyon MSP jump extends past CNNs. The contribution is a protocol, not a detector.

Contributions.

1. **Camelyon shows architecture reshuffles the logit-score ranking.** Official Kendall $W = 0.694$ ($n=8$, 7 methods, seed 42). MSP AUROC span 0.515–0.883 matches the Maha–MSP gap on ResNet-50. Copying ResNet-50’s winning logit (ReAct) onto ConvNeXt costs 0.185 AUROC. Feature-space scores stay in ranks 1–2. The seed-robust MSP jump is DenseNet-121 (0.265 ± 0.065 , 5/5); ConvNeXt and EffV2-S@224 are 4/5, MobileNet/RegNet 3/5, EffB3 2/5.
2. **Skin falsifies universality.** ISIC→PAD ($n=8$, $W=0.791$): Maha rank-1 and k NN rank-2 on all eight; MSP stays in a narrow band. Not a second Camelyon jump.
3. **AUROC is not a reliable proxy for deployment coverage.** The sign of the AUROC versus coverage@risk 10% disagreement flips between Camelyon and MIDOG (Geifman and El-Yaniv, 2019; Traub et al., 2024). On the Camelyon $n=8$

Table 1: Closest ranking-vs-backbone papers. Architecture-sensitive ranking exists on natural-image semantic OOD; the increment here is a held-fixed medical covariate split, zoo-scale W , and operational coverage.

Work	#arch.	Shift	Rank stat.	Utility
OpenOOD v1 (Yang et al., 2022)	1/set	semantic	—	AUROC
OpenOOD v1.5 (Zhang et al., 2023)	3	semantic	qualitative	AUROC
Szyc et al. (Szyc et al., 2023)	3 var.	CIFAR sem.	rank Δ	AUROC
Datko et al. (Datko et al., 2026)	per line	natural	claimed stable	AUROC
OpenMIBOOD (Gutbrod et al., 2025)	1/domain	medical	—	AUROC
Ours	8+ ViT	covar.	Kendall W	AUROC+cov.

zoo, measured on the same labeled test set, the AUROC-best and coverage@risk-best backbones differ by 0.826 coverage at a 10% risk budget; this is a retrospective comparison, not a validated selection rule (Section 4.5).

4. **A locked prediction.** Before training: ViT-B/16 Camelyon MSP ≥ 0.6947 . Observed 0.884. Held out of the 8-CNN W .

2. Related work

OpenMIBOOD (Gutbrod et al., 2025) ranks 24 post-hoc detectors with a **single frozen classifier per medical domain**. It shows that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores under that protocol. Table 1 locates the increment: we hold the dataset fixed and change the backbone.

OpenOOD v1 (Yang et al., 2022) fixes architecture on purpose. OpenOOD v1.5 (Zhang et al., 2023) compares three ImageNet models on semantic near/far OOD and reports that “different post-processor may favor different architecture,” without Kendall’s W and without a covariate hospital split. Szyc et al. (Szyc et al., 2023) show rankings change across three ResNet CIFAR implementation variants. Datko et al. (Datko et al., 2026) claim an architecture-specific winner that transfers across natural-image OOD—the opposite of a single-split concordance statistic.¹ Claros Olivares and Brockmeier (Claros Olivares and Brockmeier, 2025) rank confidence scores with AURC/AUGRC on CIFAR / TinyImageNet; they do not report a sign-flip of AUROC versus coverage@risk 10% across medical covariate domains.

OoD-Bench (Ye et al., 2022) is OOD generalization, not detection. WILDS (Koh et al., 2021) supplies Camelyon17 and iWildCam; iWildCam is a negative bound, not a third main domain.

1. This citation could not be independently verified against a primary source at the time of writing; see `refs.bib`. Confirm the original DOI/URL before submission.

3. Setup

3.1. Shifts

Camelyon17 (Bándi et al., 2019), via the WILDS (Koh et al., 2021) split: hospitals 0, 3, 4 as ID and hospital 2 as OOD (WILDS’s official OOD-test domain); founding cell, official W , $n=8$. Hospital 1 is WILDS’s official OOD-*validation* domain; we do not use it for any number in the main paper and report it only in the supplement’s held-out stain-jitter check. Skin ISIC→PAD (Codella et al., 2019; Pacheco et al., 2020): feature rank-stability without an MSP jump. MIDOG (Aubreville et al., 2023): OpenMIBOOD’s (Gutbrod et al., 2025) 1a vs. 1b+1c split and checkpoints, coverage sign-flip; official W is $n=3$. We build on OpenMIBOOD’s public MIDOG split and ResNet-50 checkpoint as an external reference. Public R50 AUROC ~ 0.59 is *theirs*; in-house R50 MSP 0.512 is *ours*. We never mix the two.

3.2. Backbones

Eight CNN-family models, ImageNet initialization, 10-epoch fine-tune, seed 42, train-frac 1.0 on Camelyon: ResNet-18/50 (He et al., 2016), DenseNet-121 (Huang et al., 2017), ConvNeXt-Tiny (Liu et al., 2022), MobileNetV3-Large (Howard et al., 2019), RegNetY-3.2GF (Radosavovic et al., 2020), EfficientNet-B3 (Tan and Le, 2019) at native 300, and EfficientNetV2-S (Tan and Le, 2021) at **224** (native 384 dropped). Standard fine-tune recipes (Adam / AdamW / SGD) follow each family’s common practice and are *not* forced-identical Adam. That confound is measured in Section 4.3. Only resolution was isolated in a controlled pair (EffB3 @224 vs. @300; EffV2-S @224 vs. @384). ViT-B/16 (Dosovitskiy et al., 2021) @224 is held out of W .

3.3. Scores and statistics

Seven post-hoc scores: MSP (Hendrycks and Gimpel, 2017), Energy (Liu et al., 2020), energy of logit-normed logits (ELogitNorm), ViM (Wang et al., 2022), ReAct (Sun et al., 2021), Mahalanobis with Ledoit–Wolf covariance (Lee et al., 2018; Ledoit and Wolf, 2004), and k NN (Sun et al., 2022). Primary concordance: **Kendall’s W per domain, never pooled**. Jump: $\text{MSP} - \text{mean}(\text{R18}, \text{R50}) \geq 0.15$. Feature drag: Maha or k NN on a new backbone below $\text{median}(\text{R18}, \text{R50}, \text{EffB3}) - 0.10$. Official Camelyon ResNet mean MSP = 0.544654; jump iff $\text{MSP} \geq 0.695$ on that split.

Coverage@risk 10% is OOD-triage: rank OOD samples by MSP (higher = more ID-like) and take the largest coverage whose classifier risk is $\leq 10\%$. ID-selective Camelyon is vacuous (ID-val acc $\approx 99.5\%$) and is not cited. AUROC and coverage@risk measure different things: AUROC is ID-vs-OD separability of the detector score alone, while coverage@risk 10% depends jointly on that ranking *and* the classifier’s own accuracy on the OOD domain, so the two need not agree, and a disagreement between them is not itself surprising without separately checking which one moved. Both quantities in Section 4.5 are computed directly on the labeled OOD test set; that makes the AUROC-vs-coverage@risk comparison there retrospective, not an executable backbone-selection procedure (Limitation 7).

We tested whether the Camelyon MSP pattern is an artifact of input resolution, temperature scaling (Guo et al., 2017), ID-only detector *selection*, a CE+SupCon recipe (Khosla

Table 2: Camelyon hospital-2 AUROC, **seed 42** (official W). Jump iff $\text{MSP} \geq 0.6947$ vs. frozen ResNet mean 0.544654. MSP range is ResNet-50 $\rightarrow$ DenseNet.

Backbone	input	Maha (rank)	k NN (rank)	MSP	seed-42 jump
ResNet-18	224	0.956 (1)	0.936 (2)	0.574	no
ResNet-50	224	0.896 (1)	0.866 (2)	0.515	no
DenseNet-121	224	0.988 (1)	0.973 (2)	0.883	yes
ConvNeXt-Tiny	224	0.928 (1)	0.916 (2)	0.846	yes
MobileNetV3-L	224	0.870 (2)	0.904 (1)	0.771	yes
RegNetY-3.2GF	224	0.964 (1)	0.956 (2)	0.784	yes
EfficientNet-B3	300	0.885 (2)	0.934 (1)	0.803	yes
EfficientNetV2-S	224	0.880 (1)	0.856 (2)	0.778	yes

Table 3: Same eight Camelyon hospital-2 MSP cells on all five seeds {42, 43, 44, 45, 46}. Jump = MSP minus that seed’s ResNet mean; bar 0.15. The last column is how many of those five seeds clear the bar. Official W is seed 42 only.

Backbone	42	43	44	45	46	jump mean $\pm$ sd	≥ 0.15
ResNet-18	0.574	0.712	0.601	0.524	0.581	—	—
ResNet-50	0.515	0.621	0.619	0.515	0.661	—	—
ResNet mean	0.545	0.667	0.610	0.520	0.621	—	—
DenseNet-121	0.883	0.865	0.868	0.845	0.826	0.265 $\pm$ 0.065	5/5
ConvNeXt-Tiny	0.846	0.799	0.803	0.764	0.785	0.207 $\pm$ 0.067	4/5
MobileNetV3-L	0.771	0.728	0.815	0.780	0.745	0.175 $\pm$ 0.081	3/5
RegNetY-3.2GF	0.784	0.778	0.836	0.830	0.724	0.198 $\pm$ 0.089	3/5
EfficientNet-B3	0.803	0.731	0.643	0.818	0.678	0.142 $\pm$ 0.125	2/5
EfficientNetV2-S@224	0.778	0.744	0.808	0.846	0.835	0.210 $\pm$ 0.089	4/5

et al., 2020), or train-time HED stain jitter. None is a clean account (Section 5 and the supplement).

4. Results

4.1. Camelyon-17 discovers the ranking shift

Official Kendall $W = \mathbf{0.694}$ ($n=8$, 7 methods). This is the only Camelyon number called “official W .” Leave-one-backbone recomputation of W ranges 0.676–0.738; that is *structural* sensitivity of the concordance statistic, not a 95% CI (supplement).

Feature methods occupy ranks 1–2 on every official backbone (Maha #1 on 6/8; k NN #1 on EffB3 and MobileNet). No backbone drags Maha/ k NN past the 0.10 feature-drag threshold. On Table 2, MSP spans 0.515–0.883; Maha 0.870–0.988; k NN 0.856–0.973. Incomplete concordance ($W=0.694$) is the logit ranks moving (Figure 2).

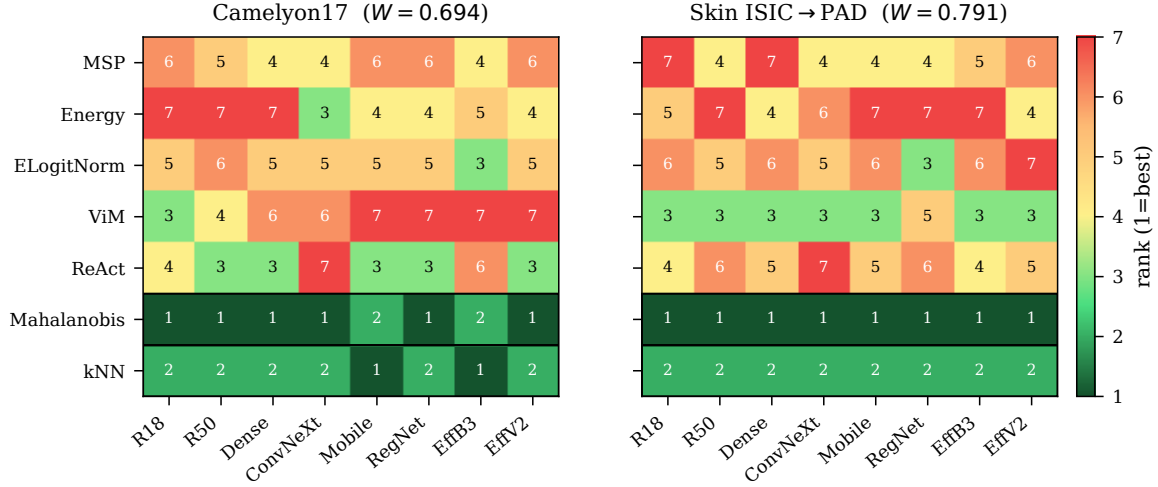


Figure 2: **Method rank by backbone** (1=best AUROC). Feature-space scores occupy ranks 1–2 on every official backbone in both domains. Logit ranks move on Camelyon ($W=0.694$) and much less on skin ($W=0.791$), where Maha is rank-1 on 8/8.

Table 3 reports the full five-seed MSP matrix: seeds verify the DenseNet MSP jump. DenseNet-121 is the confirmed seed-robust jump (5/5). ConvNeXt-Tiny and EfficientNetV2-S@224 are not seed-42 flukes (4/5) and are not promoted to 5/5. MobileNetV3 and RegNetY are 3/5; EfficientNet-B3 is 2/5 (its seed-42 jump $+0.258$ is a favorable draw relative to 0.142 ± 0.125). The seed-42 matrix still shows a jump without feature drag. The claim is narrower than “6/6 non-ResNet, confirmed.”

Copying a logit ranking is not free. ResNet-50’s best logit is ReAct (0.722); on ConvNeXt the same score is 0.692 against Energy 0.877—a 0.185 AUROC transfer cost. Feature-space transfer regret is ≤ 0.05 because Maha/ k NN stay in ranks 1–2. Architecture moves the logit leaderboard; it does not unseat the feature default.

4.2. Skin falsifies a universal MSP jump

Official $W = 0.791$ ($n=8$). Maha is rank-1 and k NN rank-2 on **all eight** (Figure 2). MSP sits in a narrow band (0.68–0.74). There is no Camelyon-style ResNet-vs-rest MSP jump. EffV2-S@224 MSP is 0.737; the native-384 cell that put k NN at rank 7 (MSP 0.823) is a resolution artifact and is not in official W . Skin is rank-stability of feature scores under a zoo: a replication of OpenMIBOOD’s feature>logit pattern *across architectures*, not an MSP-jump.

4.3. Optimizer challenges a clean architecture reading

Resolution is not the Camelyon DenseNet / seed-42 pattern. EffB3 Camelyon: 0.803 @300 $\rightarrow$ 0.787 @224 (the jump versus the official ResNet mean survives). EffV2-S Camelyon:

0.726 @384 $\rightarrow$ **0.778** @224 (the official cell; the jump is not a 384 artifact). On skin, the EffV2-S @384 k NN collapse *is* a resolution artifact and disappears at 224. These two facts are not a slogan that “resolution never matters.”

M1 — jumpers trained with ResNet Adam (lr= 10^{-4} , wd= 10^{-4} , batch 64). The four official zoo members whose standard recipe was not that Adam still jump versus the frozen official ResNet mean 0.545: ConvNeXt 0.820 (+0.275), MobileNetV3 0.737 (+0.193), RegNet 0.805 (+0.261), EffV2-S@224 0.873 (+0.328); all with id-val acc ≥ 0.994 . 4/4 clean hits. DenseNet and EffB3 already shared ResNet-Adam in the official zoo. Recipe mismatch does *not* explain the seed-42 jumps of those four members. M1 is seed 42 only; the five-seed MSP matrix is Table 3.

M2 — ResNets trained with ConvNeXt AdamW (wd=0.05). ResNet-18 MSP **0.773** (enters the 0.695 band; id-val 0.9957). ResNet-50 MSP 0.581 (stays out; id-val 0.9958). ResNet *can* jump under a non-default recipe. Architecture is not a clean variable, independent of optimizer. These two results point in opposite directions; we report both.

4.4. ViT-B/16: a locked prediction

Locked *before* training, from the seed-42 zoo rule that non-ResNet-like cells jumped: MSP ≥ 0.6947 . Linear depth/parameter/receptive-field predictors are invalid on this $n=8$ and are not quoted. Observed MSP **0.884**, Maha 0.958, k NN 0.941. The prediction holds. Not a ninth W column.

4.5. Coverage@risk: the deployment cost

Protocol: OOD-triage coverage@risk 10% with MSP, not ID-selective coverage.

On Camelyon hospital-2, the locked $n=4$ utility table has MSP AUROC order EffB3@300 (0.803) > EffB3@224 (0.787) > ResNet-18 (0.574) > ResNet-50 (0.515), but coverage@risk 10% order ResNet-50 (0.149) > ResNet-18 (0.145) > EffB3@300 (0.107) > EffB3@224 (0.013). The AUROC winner is not the coverage winner (Figure 3).

On MIDOG, official $n=3$ (in-house): MSP AUROC ResNet-50 0.512, EffB3 0.486, ResNet-18 0.438 (gaps are small). Coverage@risk 10%: EffB3 0.712, ResNet-18 0.442, ResNet-50 0.166. ResNet-50 is AUROC-highest and coverage-lowest. The permutation is *not* the Camelyon permutation.

AURC/AUGRC already assume that AUROC and coverage-based ranking can disagree (Geifman and El-Yaniv, 2019; Traub et al., 2024). The increment is the *sign* of that disagreement flipping between two medical covariate domains under the same protocol, with no rule for which backbone wins. Both quantities above are computed on the same labeled Camelyon OOD test set: this is a retrospective comparison of two different measurements of the same eight backbones, not an executable backbone-selection rule, since coverage@risk requires OOD labels that are not available at deployment time. What it shows is that AUROC order and coverage@risk order diverge by 0.826 coverage at a 10% risk budget on this zoo (DenseNet 0.078 vs. MobileNet 0.904; Figure 1b), so AUROC is not a reliable proxy for coverage@risk. A deployable estimate of coverage@risk should instead come from a held-out validation split of the target domain, not from a source-domain AU-

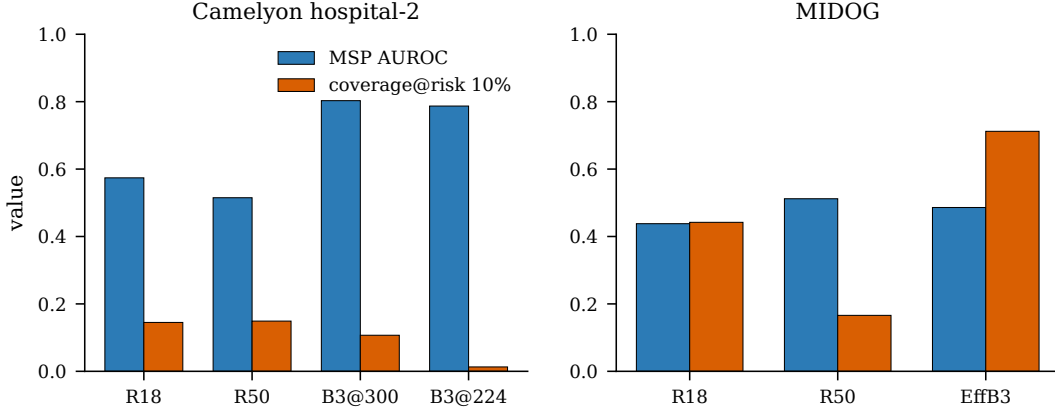


Figure 3: **AUROC vs. coverage@risk 10%**. On Camelyon the AUROC-high Efficient-Nets have the *lowest* triage coverage; on MIDOG the AUROC-high ResNet-50 is coverage-lowest. The sign of the disagreement flips.

ROC table; we flag this as an open protocol gap (Limitation 7) rather than claim a selection method here.

4.6. Frozen foundation models

Camelyon linear-probe MSP does not invert: GigaPath (Xu et al., 2024) 0.664, Phikon (Filiot et al., 2023) 0.605, UNI (Chen et al., 2024) 0.599, all >0.5 . MIDOG UNI / GigaPath MSP 0.445 / 0.470 is a MIDOG-only footnote. FM Maha ~ 0.995 –1.0 is discussion support that feature $>$ logit can survive a training-paradigm change, not a pillar.

5. Failed alternatives

Each of the following was a precommitted test. None is hidden.

Resolution (from Section 4.3). EffB3@224 MSP 0.787; EffV2-S@224 MSP 0.778. Not a pixel-size explanation of Table 2.

Temperature T^* . Correlation of T^* with the MSP jump: $r=0.268$, $p=0.52$. DenseNet–ResNet-18 MSP gap 0.308 $\rightarrow$ 0.308 after T^* . Calibration/temperature is not the mechanism.

ID-only Maha/MSP gate. Selecting a score from ID-only statistics is a *selector*, not a detector. On Camelyon it picks Maha on 2/8 backbones; mean AUROC of the gated score 0.837 versus always-Maha 0.940. The bar was matching Maha and ViM. It does not.

CE+SupCon ($\lambda=0.1$, $\tau=0.07$), ResNet-18 + DenseNet. Gap 0.308 $\rightarrow$ 0.238 (23%; pre-commit bar was 50% / ≤ 0.154). ResNet-18 MSP 0.574 $\rightarrow$ **0.515** (worse; this is the SupCon cell, not official ResNet-50 MSP 0.515). Part of the shrink is the low model degrading. Dirty; not a recipe.

HED stain jitter, an in-house MIDOG $n=8$ zoo, and a shift-type extract (covariate / task / far-OOD) are in the supplement. Joint readout: no shift-type law, no stain mitigation, no Camelyon-style MSP jump on MIDOG.

6. Discussion and limitations

The hypothesis holds on Camelyon and fails as a cross-medical MSP-jump law. Architecture reshuffles the logit-score ranking substantially—ResNet versus the rest—while feature-space scores (Mahalanobis, k NN) stay near the top across every backbone we tested; $W=0.694$ is itself a fairly high concordance, and the two score families are not equally sensitive to the backbone, so “architecture governs ranking” is true of the logit scores and not of the feature-space scores. Feature-space scores remain the stable default when the shift looks like Camelyon *and* when it does not (skin). M1 preserves the seed-42 jumps under ResNet-Adam; M2 lets ResNet-18 enter the MSP band under ConvNeXt-AdamW: architecture is the dominant axis, not a clean one. ViT-B/16 was a locked prediction, not a post-hoc ninth CNN.

The practical cost is not abstract. On the labeled Camelyon test set, the AUROC-best and coverage@risk-best backbones differ by 0.83 of OOD coverage at a 10% risk budget; whether that gap is still there when coverage@risk is estimated from a held-out validation split rather than read off the test labels is open (Limitation 7). Copying ResNet-50’s logit winner onto ConvNeXt costs 0.185 AUROC. Those are two different kinds of transfer error—one from reading a selection off an AUROC table, one from reusing a logit score across backbones—not from picking a weak detector in isolation.

Limitations. (1) **iWildCam bound.** Locked 3-backbone cell: DenseNet MSP 0.575 vs. ResNet mean 0.604, jump -0.029 . Extra five CNNs vs. that domain’s ResNet mean: ConvNeXt 0.601, MobileNet 0.558, RegNet 0.580, EffB3 0.577, EffV2-S 0.650; 0/5 jump. The Camelyon MSP-jump is absent on this non-medical covariate shift and may be biomedical-specific. iWildCam is not pooled with Camelyon W . (2) **No clean mitigation.** CE+SupCon is dirty (ResNet-18 MSP worse). HED stain jitter is dirty (DenseNet MSP $0.780 < 0.80$). The ID-only gate is a selector. (3) **MIDOG official W is $n=3$ ($W=0.857$);** the in-house $n=8$ zoo shows no MSP jump and does not replace it. (4) W is per-domain; 0.694 / 0.791 / 0.857 are not averaged. (5) No reader study: the operational claim stops at coverage@risk. Coverage@risk itself is a point estimate here; per-sample bootstrap CIs are left to the toolkit release (features are not re-extracted in this draft). (6) Architecture is confounded with its standard recipe. 5/5 seed-robustness is DenseNet only. Official W stays seed 42. Staining vs. scanner vs. tissue prep inside the hospital shift is not isolated. (7) **The coverage@risk backbone comparison (Section 4.5) is retrospective.** Both the AUROC ranking and the coverage@risk ranking there are computed on the same labeled Camelyon OOD test set; this is a diagnostic comparison of two measurements on the same backbones, not a procedure that can be run without OOD labels. A proper selection protocol would estimate coverage@risk on a held-out validation split of the target domain and report it on a disjoint test split; that experiment is specified but not yet run (decision_precommit_coverage_val_split.md). (8) **No cross-seed concordance control.** Official $W=0.694$ uses seed 42 only, and Table 3 shows ResNet-18 MSP alone ranges from 0.524 to 0.712 across five seeds. We have not computed Kendall’s W across seeds of one fixed architecture with all seven scores, which would bound how much of $W=0.694$ ’s incomplete concordance could instead be seed variance; the existing five-seed data logs MSP only and cannot answer this. That experiment is specified but not yet run (decision_precommit_cross_seed_w.md).

Code for the architecture-zoo evaluation and coverage@risk will be released, crediting OpenMIBOOD for MIDOG imglists, the public ResNet-50 checkpoint, and the 1a / cs-ID / near-OOD protocol.

7. Conclusion

Under medical covariate shift, logit-based detector scores reorder substantially with the backbone while feature-space scores stay near the top throughout: architecture governs part of the ranking, but it does not override the feature-space family’s advantage. On Camelyon that is a seed-robust DenseNet MSP jump, stable feature ranks, and a 0.83-coverage gap between the AUROC-best and coverage@risk-best backbone, both measured on the same labeled test set. Skin keeps the feature ranks and drops the MSP jump. MIDOG flips the sign of the AUROC–coverage disagreement. A pre-registered ViT hit shows the Camelyon logit instability is not a CNN-zoo accident; iWildCam does not reproduce it. The protocol is the result: rank detectors on the backbone you will deploy, and estimate that backbone’s coverage@risk on a held-out validation split of the deployment shift, not from a source-domain AUROC table.

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Appendix A. Supplementary material

Placeholder: leave-one-backbone W recomputation, HED stain-jitter check (incl. held-out hospital-1), the in-house MIDOG $n=8$ zoo, the shift-type (covariate / task / far-OOD) extract, and the full five-seed MSP figure currently live in `manuscript/cvpr2027/suppl.tex` and still need to be ported into this appendix before submission.